

5.2 Decision Theory

- We have $x \in \mathbb{R}^d$ and a target variable t .
- When you start with real world data, you don't control it. Each pair (x, t) is drawn from some statistical process.
- $(x, t) \sim p(x, t)$ (Uncertainty)
- We need $p(t|x) = \frac{p(x, t)}{p(x)}$
- Eg - Skin Lesion
 - $C_1 \rightarrow$ Absence of Cancer
 - $C_2 \rightarrow$ Presence of Cancer
- $p(C_k|x) \rightarrow$ posterior prob.

This is the decision problem.

- We are interested in the probs of the two classes, given the image, $p(C_k|x)$ - this is called the posterior probability.

Using Bayes' Theorem,
$$p(C_k|x) = \frac{p(x|C_k) p(C_k)}{p(x)}$$

- $p(C_k)$: the prior
- This is the probability of C_k , before seeing any image.
- $p(x|C_k)$: the likelihood
 - $p(x) = \sum_j p(x|C_j) p(C_j) \rightarrow$ sums to 1.

5.2.1 Misclassification Rate.

- Rule that assigns each x to one of the classes.
- We partition the input space:

$$\mathcal{R}_1, \mathcal{R}_2, \mathcal{R}_3, \dots, \mathcal{R}_K$$

- Mistake occurs when $x \in \mathcal{R}_2$ AND $t = C_1$ OR $x \in \mathcal{R}_1$ AND $t = C_2$



$$\text{So, } p(\text{mistake}) = p(x \in R_1, C_2) + p(x \in R_2, C_1) \quad (5.20)$$

$$\text{But, } p(x \in R_1, C_2) = \int_{R_1} p(x, C_2) dx$$

$$\therefore p(\text{mistake}) = \int_{R_1} p(x, C_2) dx + \int_{R_2} p(x, C_1) dx$$

We can design R_1 and R_2 in ANY way we want. The question is that how do we choose the regions so that the above integrals become as soon as possible?

- So, to minimise $p(\text{mistake})$, we should assign x to whichever class, has the minimum value of the integrand in 5-20.
- We want:
 - $p(x, C_2)$ small if we place x into R_1 .
 - $p(x, C_1)$ small if we place x into R_2 .

$$p(x, C_k) = p(C_k | x) p(x)$$

$$\text{So: } p(x, C_1) > p(x, C_2)$$

$$\iff p(C_1 | x) p(x) > p(C_2 | x) p(x)$$

$$\iff p(C_1 | x) > p(C_2 | x)$$

Therefore, we assign x to the class with higher posterior probability.

5.2.2 Expected Loss

- Up to now, we assumed (0-1) loss.
- That means:
 - all errors are equally bad.
 - That's rarely true in real life.

loss function: $L(t, a)$

So, now we choose a class with smallest expected loss.

$$E[L] = \sum_k \sum_j \int_{R_j} L_{kj} p(x, C_k) dx \quad (5.22)$$

- If the algo predicts C_j where $x \in R_k$, then the contribution to expected loss is:

$$\sum_k \sum_j \int_{x \in R_j} L_{kj} \cdot p(x, C_k) dx$$

5.2.3 The reject option.

- Posterior probabilities close to other means the classes overlap in that region of x .
- $\max_k p(C_k | x) \leq \theta$ then reject.



5.2.5 Classifier Accuracy

$$\begin{aligned} \bullet \text{ Accuracy} &= \frac{\text{correct predictions}}{\text{total predictions}} \\ &= \frac{N_{TP} + N_{TN}}{N_{TP} + N_{FP} + N_{TN} + N_{FN}} \end{aligned}$$

$$\bullet \text{ Precision} = \frac{N_{TP}}{N_{TP} + N_{FP}}$$

$$\bullet \text{ Recall} = \frac{N_{TP}}{N_{TP} + N_{FN}}$$

Joint Probability Distribution

There are two separate urns:

- Each urn has :
 - Twice as many red balls as blue balls
 - Probability of red is $\frac{2}{3}$.
 - Probability of blue is $\frac{1}{3}$.

A $\rightarrow$ outcome of urn 1.

B $\rightarrow$ outcome of urn 2.

$$P(\text{Red}) = \frac{2}{3}, \quad P(\text{Blue}) = \frac{1}{3}$$

	A = Red	A = Blue	P(B)
B = Red	$\frac{2}{3} \cdot \frac{2}{3} = \frac{4}{9}$	$\frac{1}{3} \cdot \frac{2}{3} = \frac{2}{9}$	$\frac{4}{9} + \frac{2}{9} = \frac{2}{3}$
B = Blue	$\frac{2}{3} \cdot \frac{1}{3} = \frac{2}{9}$	$\frac{1}{3} \cdot \frac{1}{3} = \frac{1}{9}$	$\frac{2}{9} + \frac{1}{9} = \frac{1}{3}$
P(A)	$\frac{4}{9} + \frac{2}{9} = \frac{2}{3}$	$\frac{2}{9} + \frac{1}{9} = \frac{1}{3}$	1

$$P(A=R | B=R) = \frac{P(A=R, B=R)}{P(B=R)}$$

$$= \frac{4/9}{2/3} = \frac{2}{3}$$

$$= P(A=R)$$

$$P(A|B) = \frac{P(B|A) \cdot P(A)}{P(B)}$$

$$\begin{aligned} P(A = \text{Blue} | B = \text{Red}) &= \frac{P(B = \text{Red} | A = \text{Blue}) \cdot P(A = \text{Blue})}{P(B = \text{Red})} \\ &= \frac{\cancel{P(B = \text{Red})} \cdot P(A = \text{Blue})}{\cancel{P(B = \text{Red})}} = P(A = \text{Blue}) \end{aligned}$$



Decision Theory :
$$P(C_k | x) = \frac{P(x | C_k) p(C_k)}{P(x)}$$

$$P(C_k | x) = \frac{P(x, C_k)}{P(x)} \quad \text{and} \quad P(x) = \sum_j P(x, C_j)$$

So,
$$P(C_k | x) = \frac{P(x, C_k)}{\sum_j P(x, C_j)}$$

$$P(C_k | x) = \frac{\overset{\text{likelihood}}{\uparrow} P(x | C_k) \cdot \overset{\text{prior}}{\text{(how common is } C_k)} P(C_k)}{P(x)}$$

By definition, joint gives the prob. of both things happening together.

prior $P(C_k)$:

$$P(C_k) = \int P(x, C_k) dx$$

likelihood $P(x | C_k)$:

$$P(x | C_k) = \frac{P(x, C_k)}{P(C_k)}$$

$$P(x) = \sum_j P(x, C_j)$$

Generative Classifiers

$$P(C_1 | x) = \frac{P(x | C_1) \cdot P(C_1)}{P(x | C_1) P(C_1) + P(x | C_2) P(C_2)}$$

Define $a = \ln \frac{P(x | C_1) P(C_1)}{P(x | C_2) P(C_2)}$

$$\Rightarrow P(C_1 | x) = \frac{1}{1 + \exp(a)} = \sigma(a) \quad \text{--- (5.40)}$$

logistic sigmoid function

• Essentially "a" is the log of the odds in favour of class C_1 , before applying sigmoid.

• If $a = 0$: both classes are equally likely $\rightarrow$ probability = 0.5

- If $a \gg 0$: strong evidence for C_1 .
- If $a \ll 0$: strong evidence for C_2 .

• $\sigma(a) = \frac{1}{1 + e^{-a}} \rightarrow$ squashes the input to $(0, 1)$

$$\sigma(-a) = 1 - \sigma(a) \quad (5.43)$$

$$\sigma(-a) = \frac{1}{1 + e^a} = \frac{e^{-a}}{e^{-a} + 1} = \frac{1 + e^{-a} - 1}{1 + e^{-a}} = 1 - \frac{1}{1 + e^{-a}} = 1 - \sigma(a)$$

Inverse of sigmoid:

$$\sigma = \sigma(a) = \frac{1}{1 + e^{-a}}$$

$$\frac{1}{\sigma} = 1 + e^{-a} \Rightarrow \frac{1}{\sigma} - 1 = e^{-a} \Rightarrow \frac{1 - \sigma}{\sigma} = e^{-a}$$

$$\Rightarrow -a = \ln\left(\frac{1 - \sigma}{\sigma}\right) \Rightarrow a = \ln\left(\frac{\sigma}{1 - \sigma}\right)$$

$$\Rightarrow a = \ln\left(\frac{p(C_1|x)}{1 - p(C_1|x)}\right)$$

$$a = \ln\left(\frac{p(C_1|x)}{p(C_2|x)}\right)$$

